Thakaa at AraGenre 2026: Definition-Guided LLM Judging with Full-Taxonomy Agreement-Gated Correction for Hierarchical Arabic Genre Classification

Hassan Alqaeri **Meshal Alamr** **Abdullah Aldahlawi**
Thakaa, Advanced Company for Information Technology
Riyadh, Saudi Arabia
{h.alqaeril, m.alamr, aldahlawi}@thakaait.net

Abstract

We report the first-place system in AraGenre 2026 (0.7352 Hierarchical Macro F1, 18 teams). The task assigns each of 27,972 hidden-test Arabic segments one of 6 broad and one of 74 specific genres, given only as English prose definitions, with the specific-label inventory *disjoint* from training. Our pipeline updates no model parameters but is not free of task-specific effort: the judge prompt carries 74 hand-written cues, and components were selected on the public leaderboard. A family-restricted, definition-guided LLM judge scores candidate genres set-wise under a hierarchically constrained decoding scheme, whose five-step pipeline reaches 0.7139. That base led on Specific Macro F1 but trailed on Broad Macro F1, so we add a correction stage: two instruction models re-predict the specific genre, each shown the *entire* 74-definition taxonomy, and a label is edited only where both agree against the base. That system scores 0.892 Broad Macro F1, alongside fewer *topic-trap* calls, where the subject a text discusses displaces the genre it instantiates. In a post-evaluation audit of instances that the base labelled as book descriptions, Religious calls fall $394 \rightarrow 233 \rightarrow 119$ as the prompt adds the type-versus-topic convention and then the full taxonomy. Refinements and an attractor drain reach 0.7352.

1 Introduction

Genre is what a text *does* rather than what it is about: the same subject appears in a contract, a news report and a textbook, and only the first is useful to someone assembling a legal corpus. Sorting documents this way is what makes an archive filterable and a corpus assemblable by register. AraGenre 2026 (El-Haj et al., 2026) frames the problem as one of generalisation: systems receive English genre definitions and limited synthetic training data, but the hidden benchmark contains noisier naturally occurring Arabic spanning Modern Standard Arabic, Classical Arabic, and multiple dialects, with *previously unseen genre–definition combinations*. Because the 74 specific test genres are disjoint from training, no specific test label can be learned from the released labelled data, and the setup pushes systems toward generalising over communicative function. We placed 1st of 18 teams (Table 2) on the official ranking metric, Hierarchical Macro F1: the mean of Broad and Specific Macro F1.

Our contributions are:

1. A hierarchically constrained, definition-guided LLM judge whose five-step base pipeline reaches 0.7139 Hierarchical Macro F1 with no model-weight updates.
2. An account of why one predictor fails the two tiers differently: broad accuracy is the family-accuracy of the specific prediction, whereas its gap with Broad Macro F1 is consistent with uneven per-class performance, so correction must be tier-specific (§4).
3. An agreement-gated correction stage in which each verifier sees the full taxonomy, with a three-arm prompt diagnostic separating an explicit type-vs-topic instruction from a visible label to move the error onto.
4. An attractor-drain step targeting the Macro F1 failure mode of heavily predicting a few generic classes; its only approximately isolated gold-scored estimate is on dev, where it was also tuned.
5. Official shared-task evaluation: 0.7352 Hierarchical Macro F1, ranked 1st of 18 teams, with the major system configurations scored as their own submissions.

The main challenges were a label inventory disjoint from training, the topic trap of §4, and classes for which our system lacks reliable written cues (song dialects, textbook levels).

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2 Related Work

Arabic and hierarchy-aware HTC. Prominent Arabic pretrained models (Antoun et al., 2020, 2021) support much work on the language, including hierarchical variants (Bouchiha et al., 2025, 2026), and a large literature models the label hierarchy explicitly (Zhu et al., 2023; Cai et al., 2024; Zeng et al., 2025; Wang et al., 2025b; du Toit and Dunaiski, 2025; Pan and Wu, 2025; Zangari et al., 2024). These closed-label supervised setups do not address AraGenre's unseen, definition-specified inventory; we enforce the hierarchy structurally, forcing broad = parent(specific).

Zero-shot HTC, LLM judging, and consensus. Prior systems reach an unseen label space through prompt ensembles (Xia et al., 2025), LLM-built prototypes (Zhang et al., 2025a) or rewritten taxonomies (Golde et al., 2026); we leave the definitions unrewritten and vary how much of the taxonomy is visible, though the base judge appends hand-written cues to each (§4). Our two-stage design retrieves candidate definitions (Chen et al., 2024) and compares them with an LLM (Sun et al., 2023; Zheng et al., 2023), with Qwen3-Embedding (Zhang et al., 2025b) placing Arabic inputs and English definitions in one multilingual space; the chain-of-thought pass follows reasoning-based HTC (Jiang et al., 2026). Ensembles of diverse predictors (Dietterich, 2000) appear in LLM form as juries, blending, mixtures of agents, debate and self-consistency (Verga et al., 2024; Jiang et al., 2023; Wang et al., 2025a; Du et al., 2024; Wang et al., 2023), which aggregate text or resample one model; we require two models to emit the *same* label.

3 Background

Each instance is `{id, text}`; systems output one `broad_genre` (of 6: Informative, Creative, Interactive, Learning, Legal, Religious) and one `specific_genre` (of 74), using exact taxonomy label names, each with an English definition. The released data are tiny and synthetic-like (train: 140 items / 7 specific genres; dev: 110 / 6), and the specific-label inventories are *100% disjoint* across train, dev and test, so no supervised label set transfers. The *hidden test* has 27,972 naturally occurring instances over 74 specific genres in 6 families: Informative (42), Learning (12), Creative (7), Religious (6), Interactive (4) and Legal (3); Informative is dominated by 20 job domains and 13 book-description topics, and Creative holds 5 song dialects. Missing predictions are counted incorrect, and Informative dominates our predictions (∼50%).

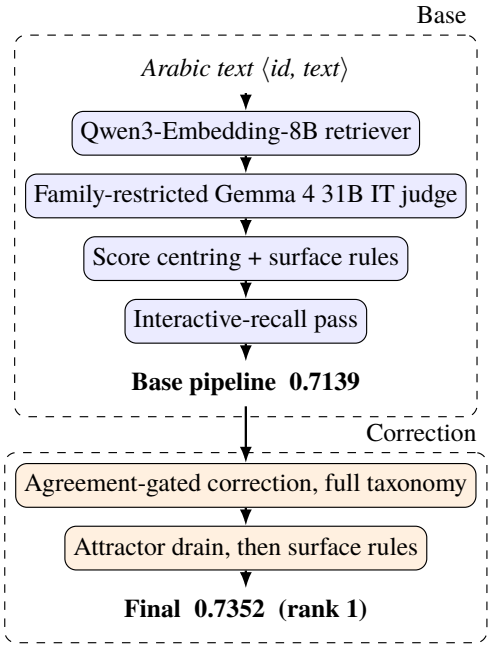


Figure 1: System overview. A Qwen3 retriever gates a broad family and a Gemma 4 31B IT judge scores its siblings setwise; score centring, surface rules and a binary Interactive-recall pass complete the five-step base pipeline (0.7139). GPT 5.6 Luna and Gemini 3.6 Flash, shown all 74 definitions, then change a label only where they agree, and an attractor drain follows (0.7352, rank 1).

4 System

4.1 Base: family-restricted definition-guided judge

The base system turns the 74-way choice into a within-family one (Figure 1): retrieval proposes candidates and a judge scores only siblings of a gated broad genre. The setwise judge is `google/gemma-4-31B-it` (Gemma Team, 2026), served through vLLM from a local copy of that checkpoint, which the endpoint reports as `/gemma-4-31b`. All three model strings are given in Appendix H. Each text passes through five steps. (i) `Qwen3-Embedding-8B` (Zhang et al., 2025b) encodes the Arabic text and the English definitions, and proposes candidate specific genres by cosine similarity. (ii) Decoding is hierarchically constrained—all siblings for small families, top-k retrieved for large topical ones—so broad = parent(specific) and broad *accuracy*

equals the family-accuracy of the specific prediction. Broad Macro F1 does not, and that gap is what the diagnosis below exploits. (iii) Following the setwise comparison of Sun et al. (2023), our judge scores each candidate 0–100 against its definition in one call, and we take the argmax after a light marginal correction (Appendix G). (iv) A few targeted surface rules override the judge only on near-unambiguous markers (Qur'anic mushaf orthography → `quran`; isnād openings → `hadith`; emoji/hashtag short posts → Interactive subtypes). These rules were assessed by convenience inspection rather than a recorded random audit, so we report no precision estimate for them. (v) Interactive remained sparsely predicted after (iv), which the prediction distribution and leaderboard feedback suggested was an imbalance, so a binary pass by the same judge over 9,978 short or marker-bearing texts reassigns 1,441 of them (Appendix A); it is in the scored base. Each definition also carries a contrastive discriminator: one Arabic line naming the signal that separates that genre from its siblings, written without instance-level gold (Appendix A). This base pipeline scores 0.7139.

Diagnosis from the scoreboard. Against the system then ahead of us, our base led on Specific Macro F1 (0.5487 vs. 0.5339) but trailed on Broad Macro F1 (0.8792 vs. 0.90), concentrating the deficit in the broad tier. Broad accuracy (0.9003) exceeded broad macro, indicating uneven per-class performance; our own predictions showed cross-family leakage, with Bible passages labelled poetry or encyclopedia entries and news labelled legal.

4.2 Correction: full-taxonomy agreement-gated re-prediction

We re-predict the specific genre with two frontier instruction models (GPT 5.6 Luna (OpenAI, 2026) and Gemini 3.6 Flash (Google, 2026)), *feeding each the entire 74-definition taxonomy* (≈2.9K tokens) grouped by broad genre, and asking for the single best specific label. The two-model design follows the ensemble argument that diverse predictors help when their errors are complementary (Dietterich, 2000; Verga et al., 2024; Wang et al., 2025a), but we never submitted either verifier alone, so agreement here is a conservatism device, not evidence of an ensemble effect. (Claude Sonnet 5 (Anthropic, 2026) was run on the diagnostic pool only, on cost grounds.)

What moves the topic-trap calls. Given only the 6 broad definitions, all three verification models produced calls consistent with classifying by subject rather than function: a blurb summarising an Islamic work *performs* the function of a book description, so the taxonomy files it under Informative while every broad-only verifier answers Religious. The full-taxonomy prompt also carries an explicit type-vs-topic instruction, so a post-evaluation three-arm ablation over the same 1,403 instances separates the two (one model, GPT 5.6 Luna, identical pool): 394 are called Religious with neither, 233 with the instruction alone (-41%) and 119 with the full taxonomy, a further 49% reduction, or 70% below the 394. The two are not interchangeable: the instruction lowers Religious calls on the book descriptions but raises them on the rest of the pool, whereas the full taxonomy lowers them on both while raising them on base-Religious instances, widening the separation from 21 to 41 points (Table 3); base-family agreement rises in five of six families (Appendix E).

Conservative application. The first model runs over all 27,972 instances, the second only over the 8,041 where the first disagrees with the base, and a correction is applied only where both name the same label. Of the 1,178 proposed cross-family moves, 460 fall in seven author-screened directions and are applied; the other 718, such as Informative→Learning, are rejected; a later rule reverts one of the 460, leaving 459 in the submitted file (Appendix I). Within-family refinements are applied wherever both models agree.

4.3 Attractor drain: redistributing to low-frequency siblings

Specific Macro F1 averages over all 74 classes, so a systematically under-predicted class has low recall and low class-level F1. This stage is partly transductive: reading our own hidden-test predictions during the evaluation phase revealed five heavily predicted *attractor* classes whose rarer siblings were almost unused, so redistribution is metric-directed toward Macro F1 and is intended to affect the per-class average more directly than accuracy; we cannot show the reassignments are semantically correct. The list comes from that distribution, not from gold. The drain is deterministic. When the base label is an attractor class and a separate chain-of-thought pass (following Wei et al., 2022) names a different sibling within the same family, the sibling replaces it;

System	Split	Hier. F1
Official baseline (embedding sim.)	dev	0.4658
Our judge (Gemma 4 31B IT)	dev	0.8794
Our judge (GPT 5.6 Luna)	dev	0.8943
Gemma 4 31B IT + correction	dev	0.8557
GPT 5.6 Luna judge + attractor drain†	dev	0.9358
Genre-general judge, no retrieval	test	0.6067
+ embedding retrieval gate	test	0.6812
+ family-restricted judging	test	0.6882
+ prompt, centring, rules, Interactive	test	**0.7139**
+ agreement-gated correction (subset)	test	0.7224
+ cross-family moves, all instances	test	0.7256
+ within-family refinements, drain	test	**0.7352**

Table 1: Results on both splits. Dev is near-degenerate (its 6 specific genres map one-to-one onto 6 broad, so all three F1 metrics coincide), so the hidden test is our primary evaluation. *Subset* corrects only instances whose broad label the base judged least secure. Provenance: test scores in Tables 1, 2 and 6 are official evaluation-phase submissions, successive rather than single-component ablations; dev scores are local. Diagnostic analyses are unscored, and the instruction-only arm (B in Table 3) ran after evaluation. † applied to the GPT 5.6 Luna judge, tuned on dev, so not held out.

the broad label never changes. The surface rules are re-applied afterwards as a guard. It moves 554 instances on the base alone and 383 in the final pipeline, where the correction runs first (Appendices I and F).

5 Experimental Setup

Data splits. Each configuration predicts automatically once fixed, but configurations were adapted between submissions from leaderboard feedback and from reading test inputs and our own outputs. An input is `{"id","text"}` with Arabic text such as مطلوب سائق باص 24 راكب في شركة في دبي جبل علي (test id 24486, "bus driver wanted, 24-seater, for a company in Dubai, Jebel Ali"); our submitted system pairs it with `broad_genre` = Informative and `specific_genre` = `drivers_and_delivery_jobs`.

Preprocessing. We apply no text normalisation; retriever inputs are capped at 512 tokens and verifier inputs at 1,500 characters, preserving heavy diacritisation (77% of texts predicted `quran`, 43% of poetry, 1% elsewhere) and emoji (28% of Interactive, under 1% elsewhere; Appendix J).

Tools, metrics and cost. Qwen3-Embedding-8B, vLLM for the judge, vendor SDKs for the verifiers; Macro F1, Weighted F1 and Accuracy at both tiers.

System	Hier	SpM	SpW	SpA	BrM	BrW	BrA
Base (ours)	.714	.549	.584	.591	.879	.900	.900
+ correction	.722	.553	.592	.598	.892	.910	.910
Final (rank 1)	**.735**	**.573**	**.613**	**.619**	**.898**	**.914**	**.914**

Table 2: Official metrics for our three scored systems on the hidden test (Sp/Br = Specific/Broad; M/W/A = Macro F1, Weighted F1, Accuracy). Sequential submissions, so each difference accompanies a configuration change, not one component: $+.013$ Broad Macro F1 with the subset correction, $+.020$ Specific Macro F1 with the final row, which combines cross-family moves, within-family refinements and the drain.

The correction makes 36,013 calls, about US$50; the base needs no paid API (Appendix H).

6 Results

6.1 Development-set results and component transfer

Dev configurations (110 examples, official gold) separate what transfers from what does not. Dev's six labels are disjoint from the test taxonomy, so discriminators, cues, rules and Interactive subtypes cannot carry over; the judge and the agreement-gated correction are re-run against the dev definitions, and of the test-specific postprocessors only the drain transfers without a test-label list. Five of the seven residual errors stem from a convention clash between splits (Appendix B). The two dev classes the drain fires on are the two most-predicted there under either judge, and it moves five labels for $+0.042$, four of them one confusion; tuned on these items, this is descriptive, not held-out, evidence. Correction does not transfer: dev has neither `*_book_description` nor scripture, and both fired corrections are wrong.

6.2 Analysis

Broad-only prompting put 157 book descriptions under Religious unanimously across all three models; the full taxonomy largely reverses that, and every alternative correction scored below the base (Appendix D).

7 Conclusion

A definition-guided judge, agreement-gated correction and attractor drain placed 1st of 18 under leaderboard selection.

Limitations

What is and is not a score. Every hidden-test *score* we report is an official evaluation-phase submission (Appendix I); we made none after the deadline. The instruction-only diagnostic arm (§4) was run after the phase closed, and the diagnostic counts analyze system predictions, including outputs from separate diagnostic runs; they are not additional officially scored results. **Validation.** The hidden test has no released gold and dev is near-degenerate, so components were selected on the public leaderboard and by manual reading; the reported progression therefore doubles as the selection signal, which risks fitting the test distribution. No hidden-test gain carries a confidence interval, so those gains are leaderboard-measured rather than statistically certified; Appendix I gives the scale of a class-level effect. Among the later correction components, only the drain showed a positive descriptive difference on dev, where it was also tuned, so its result is not held-out evidence (§6.1). **Unmeasured components.** We never ran the judge with and without the 74 discriminators and per-family cue lists, so we cannot separate their contribution from the rest of the prompt. **Partly transductive.** The marginal correction subtracts a per-genre mean computed over the whole test set (Appendix G), so the base cannot label a document in isolation, and four further choices were derived by reading our own predictions over the blind inputs: the five attractor classes (§4), the seven gated broad-move directions, the 2,635-instance verification subset (Appendix E) and the Interactive pass's pool, length thresholds, marker list and subtype defaults. No hidden label was inspected and no prediction was edited by hand, but the method is not fully inductive and none of these choices is validated against instance-level gold labels, although leaderboard scores provided an indirect gold-based selection signal. **An untested comparison.** We ran GPT 5.6 Luna over the full test set but never submitted it as a system, so the claim that a cheap base judge with conservative correction beats direct full-taxonomy prediction is untested. **Cost and ceilings.** Correction cost scales with test size, and our cue inventory covers few song-dialect predictions, so it provides limited evidence for those distinctions.

Ethics Statement

This work uses only the publicly released AraGenre data and collects nothing new. In line with the shared-task rules all predictions are produced automatically: no submitted label was hand-edited and we did not reverse-engineer the hidden evaluation labels. We did read system outputs and their texts to design the automatic rules, but that is inspection of our own outputs, not annotation of the evaluation set, and no such judgement enters a submission except through a deterministic rule.

The taxonomy spans religiously and culturally sensitive categories, which we treat descriptively as text-type labels: a misclassification carries no judgement on the authenticity, orthodoxy or quality of a text, and models trained on skewed corpora may under-serve lower-resourced dialects. Two misuses would concern us. Dialect is not provenance: pointed at ordinary prose, the classifier reads as a guess at where a writer is from, and we have no evidence it supports that use. And because the labels separate religious from informative text, genre output could be used to route or suppress material by register rather than by content. The system assigns text types, it does not assess them, and a 0.57 Specific Macro F1 is not a basis for an automated decision about a document or its author without a person in the loop.

Our correction stage sends Arabic text to third-party hosted APIs; the inputs are already-public benchmark data, but the same pipeline should not be applied to private text without review. Reliance on hosted models whose cost and behaviour may change limits equitable reproducibility; the base retriever-plus-open-judge pipeline is the self-hostable alternative.

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A Reproducibility

Official submission. The results we report are those of our final submission, evaluated 1 August 2026, which scored 0.7352 Hierarchical Macro F1 (Specific Macro 0.5725, Weighted 0.6125, Accuracy 0.6189; Broad Macro 0.8979, Weighted 0.9142, Accuracy 0.9138) over 27,972 of 27,972 gold instances with no missing prediction, ranking 1st of the 18 teams in the organisers' final standings. Every hidden-test *metric score* we report for a system configuration comes from an official submitted file rather than a local estimate. Diagnostic counts analyze system predictions, including outputs from separate diagnostic runs; they are not additional officially scored results. The full record is in Appendix I. There are no model-weight updates at any stage; task-specific information enters through the hand-written discriminators, cue and dialect-marker lists, surface rules, dev-tuned prompts, the attractor set, the accepted broad-move directions, the verification subset, the centring constant, the cached broad gate, the Interactive-recall pass and leaderboard-guided component selection.

Data: we use only the official AraGenre inputs and definition files, and no external labelled data. The consensus verifiers receive the 74 definitions verbatim. The judge does not: by default it appends a one-line Arabic discriminator to each definition and prepends per-family cue lists, both author-written from the released definitions and Arabic linguistic knowledge, and both naming genres explicitly (`scripts/discriminators.py`, `scripts/run_family_judge.py`). The surface rules name a further seven labels. No hidden-test gold label or annotation informs any of it, but the pipeline is not label-agnostic and we do not claim it is.

Retrieval coverage. The hidden test has no gold, so we cannot report recall against true labels. We can report how often retrieval surfaced the label the system itself ended up choosing: the base label lies in the retrieved top-20 for 98.4% of instances, and in the top-6 of that same ranking over all 74 candidates for 93.1% of the instances the base assigns to the two large families, whose judge shortlists are cut from it. The judge's output before the surface rules and the Interactive reassignment necessarily

lies within its scored shortlist, so coverage of that shortlist carries no information. The final label lies in the top-20 for 97.2%; the remaining 2.8% are labels the correction stage, the rules, the Interactive pass or the drain introduced from outside the retrieved set. These are coverage figures for our own predictions, not accuracy.

Provenance of the hand-built parts. Nothing below was tuned on held-out gold. The 74 discriminators and per-family cue lists were written by one author from the released definitions and Arabic linguistic knowledge, without access to instance-level gold labels. Their file was created in a single write on 28 July 2026 and never modified afterwards, so they were composed after the hidden-test inputs were released and after our first four scored submissions, but were fixed two days before the configuration that uses them was scored and could not have been revised in response to any later leaderboard result. A contemporaneous design note records the rule followed—no class frequencies or priors were to be injected—and contains only definitional and dialect-marker material, no quoted test instances. We did not log whether hidden-test texts, our own predictions, aggregate prediction patterns or the first four leaderboard results were consulted during cue composition, so we cannot rule out influence from any of them. The five attractor classes, the seven accepted broad-move directions and the 2,635-instance verification subset were chosen by reading our own predictions over the released test inputs. The centring constant α was set by hand (Appendix G), and the broad gate is an earlier scored submission of ours reused verbatim.

Models: the base retriever is Qwen3-Embedding-8B (4-bit NF4 weights with fp16 compute, 4096-dim, cosine over L2-normalised vectors, sequence length capped at 512 tokens). The judge is a Gemma-family model (`gemma-4-31b`) served via vLLM, scoring candidate definitions setwise on a 0–100 scale. Consensus verification uses GPT 5.6 Luna and Gemini 3.6 Flash, each queried with the full-taxonomy prompt.

Settings: the judge decodes at temperature 0 with seed 42; the Gemini verifier is called with `temperature=0`; the GPT verifier uses `reasoning_effort=low` and no temperature set. We record the parameters we sent; whether a hosted model honours them is outside our control, and we do not treat `temperature=0` as guaranteeing deterministic output. The retriever caches the top 20 specific genres per text; the judge scores every sibling in the four small families and the top 6 retrieved elsewhere, in a single setwise pass, with marginal correction α=0.75 (Appendix G).

Pipeline and code: code, prompts, text-free cached model outputs, the exact submitted JSON, and a publication-clean ZIP containing it are available at https://github.com/Hassn11q/thakaa-aragenre-2026. Every path below is relative to the repository root. The submitted stages are retrieval (`scripts/run_retrieval.py`), the family-restricted judge (`scripts/run_family_judge.py`), the surface rules (`scripts/apply_rules.py`), the full-taxonomy verifiers (`scripts/run_full_taxonomy_verifiers.py`), the consensus and drain assembly (`scripts/build_final.py`), and the submission builder (`scripts/build_submission.py`), which enforces broad=parent(specific) and validates that all 27,972 IDs are present with exact labels.

Scope of reproducibility. The final submission can be reconstructed exactly from the cached outputs, but the base pipeline cannot be regenerated end to end. Two inputs to the trap experiment illustrate the gap: the 2,635-instance verification pool, selected during the evaluation phase as the instances whose broad label we judged least secure, and the broad-only prompt behind arm A. The pool IDs and all three text-free cached arms are released, and `scripts/rebuild_prompt_diagnostics.py` rebuilds the reported counts offline, but neither selection can be re-derived from the code.

Two cached inputs. The broad gate the judge restricts to is the scored 0.6812 system, an earlier retrieval-gated system we submitted and scored, reused verbatim rather than regenerated by the released code; and the base predictions also carry an Interactive-recall reassignment applied during the evaluation phase by a separate automatic judge pass (`scripts/run_interactive_recall.py`), which is released but not wired into the offline reconstruction; rerunning the judge over the same candidate sets reproduces 93.5% of the base specific labels and 97.1% of the broad labels. Both submitted caches are released, so `run_reproduce.sh` rebuilds the submitted predictions exactly (27,972/27,972) without hidden-test text, API access or a GPU, but the base stage is a cached input rather than something the repository regenerates end to end.

Caveats: the judge endpoint's serving precision was not recorded (Appendix H), so the base stage is not bit-reproducible even with a GPU; the hosted API models are versioned by release date and are not guaranteed bit-reproducible across dates; the consensus stage incurs API cost proportional to the number of verified instances (on the order of tens of US dollars for this test set). The base pipeline runs on a single CUDA GPU (embeddings) plus a remote vLLM endpoint (judge).

B Annotation-Convention Divergence Between Splits

Five of the seven residual dev errors are texts that analyse religion from the outside, e.g. تشير بعض الدراسات إلى أن أساليب الوعظ تختلف بين المجتمعات (*tushīru ba'ḍu al-dirāsāti ilā anna asālība al-wa'ẓi takhtalifu bayna al-mujtama'āt*, "some studies indicate that preaching styles differ between societies"). Our system labels these `Analytical Reports` (Informative); the dev gold label is `Religious Commentary` (Religious).

The released test taxonomy points the other way: `islamic_book_description` ("texts summarising Islamic or religious books") is Informative and our correction stage moves such predictions out of Religious, consistently with the released test definition. The examples suggest different boundary choices across the disjoint dev and test taxonomies, although the differing label inventories prevent a direct annotation-consistency comparison. A system following the released test taxonomy is therefore at a disadvantage on these dev examples, an observed tradeoff rather than a formal bound. It is also consistent with our central claim: the convention is carried by the definitions, which is why supplying the full taxonomy, rather than the broad labels alone, changes model behaviour so sharply.

C Prediction Audit Examples

Three recurring prediction patterns appear in our audit. (i) *Cross-family topic leaks*: a description *of* a religious book (يهدف هذا الكتاب إلى قراءة سيرة الرسول, *yahdifu hādhā al-kitābu ilā qirā'ati sīrati al-rasūl*, "this book aims at a reading of the Prophet's biography", id 10041) is `islamic_book_description` (Informative), which both our base and final systems label that way, consistent with the released definition, but which all three verifiers call Religious under the broad-only prompt; scripture leaks the other way (هؤلاء امراء بني عيسو, *hā'ulā'i umarā'u banī 'Īsū*, "these are the chiefs of the sons of Esau", Genesis 36, id 10798), where the base says `history_encyclopedia` and consensus changes it to `bible`. (ii) *Heavy attractor prediction*: long prose absorbed into `egyptian_song_lyrics`. (iii) *Low cue coverage*: for each of the five song-lyric classes, 87–99% of the texts we assign to it carry no marker from our cue inventory (the fixed cue inventory is embedded in `scripts/run_family_judge.py`). This measures the coverage of our inventory, not an intrinsic ceiling: it shows our system lacks reliable surface cues for these distinctions, not that no written signal exists.

D Negative Results and Unscored Explorations

Two further paths were built but never scored, so we report no outcome for them: LLM-enriched definitions (a sibling contrast and two synthetic exemplars per genre) and a transductive label-propagation pass over the test embeddings. We record them because the release includes their scripts; the deployed judge keeps the definitions verbatim and adds only the hand-written cues. A submitted configuration whose broad label came from a *direct* six-way judge, rather than from the specific prediction, scored 0.6206; the submitted configuration that scored all 74 candidates without retrieval pruning scored 0.6517. Neither isolates a single component. For correction, plain self-consistency (0.7134), entropic optimal transport (0.7115), uniform-prior calibration (0.7124) and an unrestricted chain-of-thought pass (0.7131) all regressed below the 0.7139 base, whereas full-context consensus improved it. Aarab (2026) reports rerankers as the strongest zero-shot classifiers on their benchmark, but the cross-lingual rerankers we tried did not help on this task; regex rules did not improve the 0.6882 configuration, and the off-the-shelf dialect classifiers we tried did not expose the Iraqi/Sudanese distinction that NADI-style country-level tasks (Abdul-Mageed et al., 2021) define.

E Full-Context vs. Broad-Only Prompting

Both verifiers were run over the same verification pool ($n = 2{,}635$) once with only the 6 broad definitions and once with all 74 specific definitions. That pool was selected during the evaluation phase as the instances whose broad label we

judged least secure, so it is an enriched diagnostic sample: rates measured on it are not representative of all 27,972. Because the full-taxonomy prompt also carries an explicit type-vs-topic instruction, we added a third arm, broad definitions only *plus* that instruction, to separate the two factors. On the 1,403 book-description instances GPT 5.6 Luna calls 394 of them Religious with neither, 233 with the instruction alone and 119 with the full taxonomy; `scripts/rebuild_prompt_diagnostics.py` rebuilds these counts from cached verifier outputs with no API access.

Table 3 splits the same three arms over the whole pool. Two things follow. First, the reduction is not uniform across subsets: Religious calls decrease on base-non-Religious instances but increase on base-Religious instances. Second, the instruction and the taxonomy are not interchangeable. The instruction alone reduces Religious calls on book descriptions; adding the full taxonomy produces a further reduction and a different pattern across the remaining subsets. The caveat of Table 4 applies here too: the row labels come from the base system, so these are agreement counts, not accuracy.

Religious calls	A	B	C
book descriptions (n=1,403)	394	233	**119**
other non-Religious (n=1,169)	121	152	**61**
base-Religious (n=63)	26	28	**30**
separation (pp)	21	29	**41**

Table 3: The three arms over the full verification pool (GPT 5.6 Luna). A: broad definitions only. B: + type-vs-topic instruction. C: + full 74-definition taxonomy. Rows group instances by their base label; the first two should fall and the third should rise, and only C does both. Separation is the Religious-call rate on base-Religious instances minus the rate on the other 2,572. Rebuilt by `scripts/rebuild_prompt_diagnostics.py`.

Table 4 reports how often GPT 5.6 Luna's broad verdict still agrees with the base system's family, one model across the two original prompts. Broad-only prompting shows substantially lower base-family agreement on families whose members are *about* another family's subject matter (Learning, Informative); full-taxonomy prompting raises that agreement. Requiring both verifiers to agree, joint Religious assignments fall from 134 under broad-only prompting to 37 under the full taxonomy, down by 72%. That comparison is restricted to the 548 book descriptions for which the second verifier has a call under both prompts, since the deployed full-taxonomy run queries it only on the instances where the first verifier disagrees with the base; on all 1,403, the first verifier alone falls from 394 to 126. That 126 is the deployed all-instance run scored on this subset, not arm C, which is a separate pool-matched run of the same prompt. The two were issued 26 minutes apart on 31 July 2026 with the same model string and the same instances. We did not record the model build returned by either call, so the seven-prediction gap cannot be attributed to a specific cause: run-to-run variation and a service-side model change are both consistent with it. All three broad-only runs are included, and `scripts/rebuild_prompt_diagnostics.py` reproduces these counts and the 157 above without API access.

Base family	n	broad-only	full-taxonomy
Informative	1682	49%	**78%**
Learning	305	18%	**74%**
Interactive	153	46%	**72%**
Legal	281	53%	**56%**
Creative	151	50%	48%
Religious	63	41%	**48%**

Table 4: Share of instances where GPT 5.6 Luna's broad verdict matches the base family, by prompt context. This measures *agreement with the base system*, not accuracy: the hidden test has no released gold, so we can show that full context changes verifier behaviour and that the change is consistent with the taxonomy's own conventions, but we cannot certify it as an accuracy gain. Dev cannot serve as the gold-anchored counterpart here: it contains no book descriptions and no scripture, so the trap this table measures has no instances there (§6.1).

F Attractor Drain

Relative to the base, the final system changes 459 broad and 2,890 specific labels (10.3% of instances). The drain is deterministic: it fires when the base label is an attractor class and the chain-of-thought pass names a different sibling inside the same broad family (Table 5). No score threshold or entropy criterion is used, and broad is held fixed by construction.

G Marginal Score Centring

This is score centring, not probabilistic calibration: we subtract a scaled per-genre mean from each candidate's score, $s'(y) = s(y) - \alpha\,\bar{s}(y)$, where $\bar{s}(y)$ is the mean score genre y received over the instances

Class	base	moved
literature_fiction_bk_desc.	598	208 (35%)
egyptian_song_lyrics	979	212 (22%)
msa_poetry	608	52 (9%)
culture_encyclopedia	455	41 (9%)
history_encyclopedia	632	41 (6%)

Table 5: Attractor classes and how many predictions the drain reassigns to within-family siblings when applied to the base predictions alone. In the submitted pipeline consensus is applied first and the drain fires on 383 of these instances.

in which y appears in the scored candidate set, recomputed independently for each split, and α=0.75, chosen on the released training labels. It damps genres the judge rates generously everywhere, which is what leaves rare siblings seldom predicted. The submitted base invocation was not logged. A contemporaneous report records α=0.75, while the corresponding experiment records selecting α over the grid $\{0, 0.25, 0.5, 0.75, 1\}$ record other values, including 0.5 and 1.0. Two cached prediction files, the chain-of-thought pass (`cot_predictions.json`) and an earlier base rebuild (`early_base.json`), reproduce at α=0.75 and at no other value on that grid. We therefore report 0.75 as a reconstruction rather than a directly verified invocation setting. Centring runs before the argmax and never changes the broad family. Because $\bar{s}(y)$ is a batch statistic, this stage is transductive: a document's label depends on the set it is scored with. We tried two alternatives instead of it, uniform-prior calibration and entropic optimal transport, and both scored below the base (Appendix D).

H Hyperparameters and Prompts

Retriever: `Qwen/Qwen3-Embedding-8B` loaded in 4-bit NF4 with double quantisation and fp16 compute. These are not reconstructed: the released retrieval script (`scripts/run_retrieval.py`) records these settings; only the cosine ranking is consumed, and we did not measure the effect of NF4 quantisation on retrieval rankings, 4096-dim, L2-normalised, `max_seq_length` 512, the top 20 specific genres cached per text. Only the 74 *specific* definitions are embedded, with no prefix; the broad definitions are not. The Arabic texts alone carry the Qwen3 query instruction `Instruct: Given an Arabic text, retrieve the genre definition that best describes it.` followed by `Query:`, and we recorded no checkpoint revision hash for either the retriever or the judge. The judge scores every sibling in the four small families (Creative 7, Religious 6, Interactive 4, Legal 3) and the top 6 retrieved in-family candidates in the two large ones.

Judge. A local copy of `google/gemma-4-31B-it`, served via vLLM and reported by the endpoint as `/gemma-4-31b`, served with `max_model_len` 32,768 rather than the checkpoint's full context; setwise 0–100 scoring in a single pass, temperature 0, seed 42, up to 3 attempts with JSON-regex fallback, marginal correction α=0.75. We did not record the serving precision of the judge endpoint. The retriever precision is given above.

Interactive-recall and drain passes. Both reuse the judge endpoint and its settings (temperature 0, seed 42). The Interactive pass is binary. Its pool is every text at most 400 characters long whose base label is neither Interactive nor `quran`/`hadith`, and which is either under 220 characters or carries an emoji, a hashtag or one of a fixed dialectal-pronoun list; both thresholds and the marker list were set by reading our own predictions during the evaluation phase, not from gold. It asks for a JSON object carrying an interactive flag and a subtype ($\leq$40 tokens). A negative verdict keeps the base label; a positive one takes the returned subtype, falling back to `youtube_comments` when the subtype is not one of the four Interactive classes. Replies that yield no parsable object are retried up to three times, and the run recorded no instance that exhausted them. It moved 1,441 predictions (`scripts/run_interactive_recall.py`). An earlier Interactive-subtype configuration was scored (0.6942, Table 6), but the pass as incorporated into the 0.7139 base was never scored as a matched isolated addition. The drain reads the chain-of-thought pass, which scores the same candidate sets as the base judge but emits a short rationale before the scores; unparsable replies keep the base label. Neither pass is wired into `run_reproduce.sh`, which consumes their cached outputs.

Diagnostic-only verifier. Claude Sonnet 5 (`claude-sonnet-5`, $\leq$16 completion tokens, temperature not set, same broad-only prompt as the other two) was run for the broad-only diagnostic in §6.2 and appears in no submitted configuration.

Consensus verifiers. GPT 5.6 Luna (`gpt-5.6-luna`; `reasoning_effort=low`, ≤500 completion tokens) and Gemini 3.6 Flash (`gemini-3.6-flash`; `temperature=0`, ≤400 output tokens), each given the full 74-definition taxonomy (2,884 tokens under `o200k_base`, giving a mean prompt of 3,034 tokens and a maximum of 5,185) in the system prompt, and the text truncated to 1500 characters. The judge prompt asks for a JSON array of integer scores over the candidate definitions; the consensus prompt asks for a single exact `specific_genre` identifier and states that a factual description *of* a book is a `*_book_description`, not the book's subject genre.

Software. The retriever uses `torch`, `transformers`, `sentence-transformers` and `bitsandbytes`; the local judge runs under vLLM; the verifiers use the `openai` and `google-genai` SDKs. The code lists minimum versions rather than exact pins.

Cost. The $50 in §5 is the amount billed for the 36,013 calls at July 2026 prices; we did not log per-request token usage, so we cannot restate it as a rate.

Truncation and malformed outputs. The retriever's 512-token cap binds on 3,365 of 27,972 instances (12.0%; median length 127 tokens, 90th percentile 587, maximum 3,346, measured post hoc with the Qwen3 tokenizer), and the 1,500-character verifier cap binds on 2,987 (10.7%); no text reaches the 8,000-character pre-truncation cap. The first verifier returned an unparsable label on 2 of its 27,972 calls and the second on 33 of its 8,041; those instances keep the base label.

Throughput. A consensus call takes 2.0–2.8 s. With the released default of 20 concurrent workers the 8,041-instance Gemini pass took 19 min of wall clock; the full GPT pass over 27,972 instances was run at higher concurrency.

I Official Submission Record

Every hidden-test metric score we report for a system configuration comes from an official submitted file (Table 6), so the progression is traceable to scored submissions rather than to local approximations. Diagnostic counts analyze system predictions, including outputs from separate diagnostic runs; they are not additional officially scored results.

System	Hier. F1
genre-general judge, no retrieval	0.6067
+ embedding retrieval gate	0.6812
+ family-restricted judging	0.6882
direct six-way broad judge	0.6206
unpruned 74-way scoring	0.6517
+ surface rules only	0.6881
rule variant	0.6891
prompt-ensemble blend	0.6883
Interactive-subtype judging	0.6942
hybrid judge variant	0.7128
base system	0.7139
uniform-prior calibration	0.7124
self-consistency (K=3)	0.7134
entropic optimal transport	0.7115
chain-of-thought applied wholesale	0.7131
+ agreement-gated corr. (subset)	0.7224
+ cross-family moves, all inst.	0.7256
+ within-family refs, drain	**0.7352**

Table 6: Official submissions behind Table 1 and Appendix D. The final system produced 27,972 predictions over 27,972 gold instances, none missing. We used 19 of our 26 permitted submissions; the 18 scored on the test phase are all listed here. Submissions were made sequentially over four days, so the progression is a record of scored configurations rather than a single controlled sweep.

The scale of a stage gain. Every specific class carries the same $1/74$ weight in Specific Macro F1 regardless of how often it is predicted, so moving one class's F1 from 0 to 1 shifts Specific Macro F1 by 0.0135 and Hierarchical Macro F1 by 0.0068. Measured against that unit the individual stages in Table 1 are small: the correction on the subset is worth about 1.25 class-level effects, extending the cross-family moves about 0.47, and the within-family refinements with the drain about 1.41. The full base-to-final gain is about 3.1 and the margin over the second-placed system about 2.7. These are coarse yardsticks, not significance tests.

Two readings this record supports. The 0.6881 row isolates the surface rules on top of the 0.6882 row: they rewrite 32 labels (23 `quran`, 9 `hadith`) and leave the score at 0.6881 against 0.6882. In the 0.6882 configuration, adding the rules alone changed the score to 0.6881. Because the later 0.7139 submission jointly changed the prompt, cues, score centring, rules and the Interactive-recall pass, that gain cannot be decomposed from the available submissions. They were kept for a different job: in the final pipeline they fire on 1,087

instances and agree with the base label on every one, so their whole net effect is to restore 75 labels that consensus or the drain had moved away (56 `youtube_comments`, 19 `quran`). Their observed mechanical effect in the final assembly is to restore those 75 labels; we have no scored submission of the final system without them. The chain-of-thought row applies that pass to every instance and scores 0.7131, below the 0.7139 base; the same pass restricted to attractor classes is what the drain uses. We have no drain-only hidden-test score, so we cannot attribute a gain to the restriction.

How the directions were screened. For each candidate direction we read a sample of the moves the two verifiers agreed on and asked whether they showed a recurring misfiling that the released definitions themselves resolve—dialectal comments filed as book descriptions, wire news filed as resolutions or contracts, long prose filed as song lyrics, scripture filed as fiction or encyclopedia entries. Directions where the agreed moves instead looked like a shared topic bias were rejected, the clearest being Informative→Learning, which over-fires student-writing on academic prose. The criterion is a reading judgement, not a measurement; the accepted set is fixed in `scripts/build_final.py` with a one-line rationale per direction. One author did the screening, and a contemporaneous note shows the practice: the Interactive direction was diagnosed by reading test texts, with four instances quoted verbatim. We did not record how many moves were read per direction or whether they were sampled at random, so the scale of that inspection is not recoverable.

Gated broad moves. Of the 3,300 instances where both verifiers agree against the base, 2,122 stay inside the base family and are applied as refinements. Of the 1,178 that would cross families, 460 fall in an author-screened direction and are applied: Informative→Interactive 195, Legal→Informative 80, Creative→Religious 58, Informative→Religious 58, Creative→Informative 44, Interactive→Religious 20, Legal→Interactive 5; the final file differs from the base on 459 broad labels, one fewer, because the closing surface-rule pass restores one. The remaining 718 are rejected, the largest groups being Informative→Learning (219), Creative→Interactive (122) and Learning→Informative (99).

Preserved rule defects. Three regexes in `scripts/apply_rules.py` do not match their intent: the Qur'anic class also admits U+0671, the opinion class lets a bare variation selector match, and one character class contains a literal `|`. They ran as written when the submission was produced, so we keep them verbatim and document them in the code; correcting all three changes 22 of the 27,972 final predictions.

J Orthographic Signals

Share of each group carrying heavy diacritisation (a mark on over 10% of its Arabic characters) or at least one emoji, over the final predictions: `quran` 77%/0% (n=691), poetry (Classical and MSA) 43%/0% (n=1,606), `hadith` 1%/1% (n=647), book descriptions 1%/0% (n=3,916), the whole Interactive family 0%/28% (n=1,978; `twitter_posts` 57%, `youtube_comments` 30%, `app_reviews` 11%, `book_reviews` 0%). Both are properties of the text, so no gold is needed; the released submission and official test text reproduce these descriptive counts.